

RF SYSTEMS ENGINEER

Professional Summary

Results-driven RF Systems Engineer with over eight years of progressive experience in radio frequency hardware design, satellite communications architecture, and manufacturing validation. Proven track record of delivering complex multi-disciplinary projects on schedule and under budget. Adept at collaborating with cross-functional teams to translate system-level requirements into manufacturable hardware solutions. Committed to driving continuous improvement in design quality and production yield.

Core Competencies

RF and Microwave Design | Satellite Communication Systems | Antenna and Filter Design Design for Manufacturability (DFM) | EVT/DVT/PVT Validation | Root Cause Analysis Signal Integrity | Link Budget Analysis | Cross-Functional Leadership | Supplier Management

Technical Skills

Simulation Software: Ansoft Designer, AWR Microwave Office, ADS, HFSS Test Instrumentation: Spectrum Analyzer, Vector Network Analyzer, Signal Analyzer, Logic Analyzer Programming Languages: Python, C/C++, MATLAB, LabVIEW Circuit and PCB Design: SPICE, NI Multisim, Altium Designer, Eagle

Professional Experience

Senior RF Systems Engineer CompanyName | 06/2018 to Current - Led the design and validation of RF front-end subsystems for next-generation satellite payloads, improving overall system noise figure by 18 percent. - Developed comprehensive DFM procedures and design checklists for antennas, receivers, filters, and amplifiers, reducing supplier design iterations by 30 percent. - Directed EVT, DVT, and PVT validation campaigns across three concurrent programs, achieving 100 percent on-time completion of qualification milestones. - Established cascade RF performance prediction models covering noise figure, gain, IP3, and spurious response, improving correlation between prediction and measurement to within 0.5 dB. - Mentored four junior engineers in RF measurement methodology and design review practice.

RF Design Engineer CompanyName | 08/2015 to 05/2018 - Executed root cause analysis on field return units, identifying a solder reflow defect that reduced warranty claims by 22 percent year over year. - Collaborated with contract manufacturers to implement automated test sequences, decreasing per-unit test time by 40 percent. - Authored technical design specifications and acceptance criteria for six supplier hardware deliverables. - Conducted on-site supplier audits and design for manufacturability reviews across two international facilities.

Systems Data Analyst CompanyName | 07/2013 to 07/2015 - Designed and maintained SQL-based data collection pipelines supporting laboratory measurement campaigns, improving data completeness from 82 percent to 99 percent. - Developed statistical analysis workflows in Python and SAS to identify yield drivers across production lots. - Produced executive-level reporting dashboards summarizing test results for program leadership.

Education

Master of Science, Electrical and Computer Engineering CompanyName University | 2013 GPA: 3.9 / 4.0

Bachelor of Science, Electrical and Computer Engineering CompanyName University | 2011 GPA: 3.2 / 4.0

Certifications and Professional Development

Certified LabVIEW Associate Developer Six Sigma Green Belt IPC-A-610 Acceptability of Electronic Assemblies